

CSCI 205: Five Nights In Dana

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Aiden Cherniske, Faris Lutfi



▪ Bucknell
UNIVERSITY

About the Team



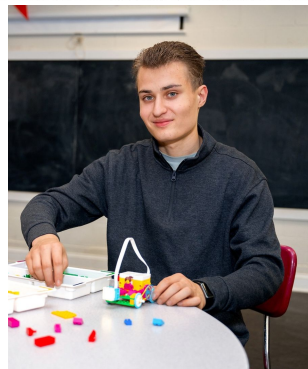
Mason Barlow

Major: Junior Computer Engineering Major

Role: Developer

Tasks:

- Assisted Systems Implementation
- Technical Writing
- Map Design/Art Design



Aiden Cherniske

Major: Junior Computer Engineering Major

Role: Product Owner

Tasks:

- Project Planning and Scaffold
- Main AI Behavior
- Core System Development
- Core System Integration



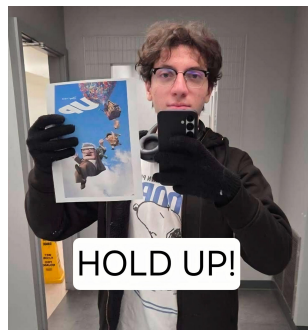
Tyler Seitz

Major: Junior Computer Engineering Major

Role: Scrum Master

Tasks:

- UI development
- Lead Scene Builder Design
- Core System Integration
- Scrum Duties



Faris Lutfi

Major: Sophomore Computer Engineering Major

Role: Developer

Tasks:

- Asset Generation / Sound Design
- Technical Writing

Five Nights at Dana

Overview:

- Our game is a point-and-click style game based on Five Nights at Freddy's.

Story:

- You play as Professor Lily grading late at night: last through the night by monitoring camera feeds, reacting to movement across the building, and managing limited resources before the students realize you are still available for late night questions.



User Personas

Completionist

Narrative: Achievements and unlockables are his primary motivation for coming back after beating the main story.

Motivation: Make a game which is unique on every playthrough and with enough variation for multiple plays

Lore Hunter

Narrative: He is drawn to games with deep, hidden storytelling.

Motivation: Develop a game with a fun backstory and lore to keep players engaged and amused

Game Design Appreciator

Narrative: Jen has a deep appreciation for the craft behind game development.

Motivation: Create an immersive experience with sounds, art, and gameplay which can be appreciated by the player

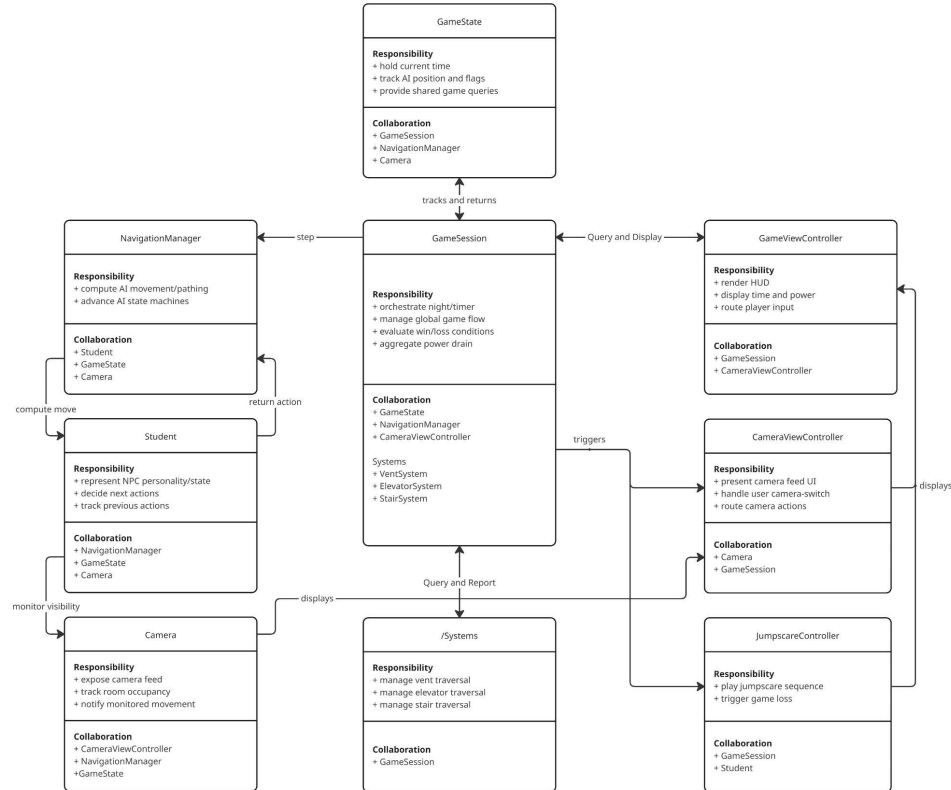
Casual Gamer

Narrative: She appreciates clear tutorials and forgiving early levels that let her learn the mechanics gradually

Motivation: Implement a scaling game with clear instructions which is accessible to new players and veterans alike

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UML Diagram



Code Design

Student Class

Internal State Management (Encapsulation):

- The system keeps internal fields private
- Stores name and personality
- Calls update() every simulation tick

Behavior/Path Abstraction via Enumerations:

- Complexity is hidden within the Personality and Path Type enumerators.
- Designates paths students prefer to take and allows for individual tuning



Student

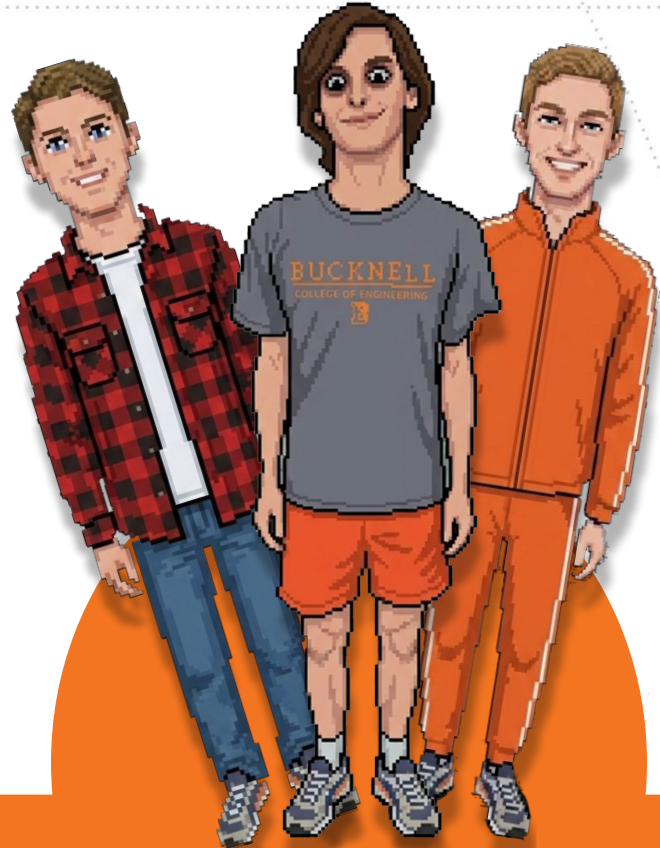
Movement

The core movement decision logic will:

- Rolls 1-20 and moves if roll $\leq$ aiLevel
- aiLevel increases as the game progresses to scale difficulty
- Calls doMoveStep() to compute the next location via NavigationManager class

Memory

- rememberLocation(Location) adds a location to a FIFO deque to remember recent locations
- Used to pushBack(int steps) a student rewinding their location (also can be pushed random number of steps)



Special AI Types

Runner

- When not checked, Runner will become restless and come sprinting down the hall
- Timer resets when camera in Computer Lab is checked

Instant

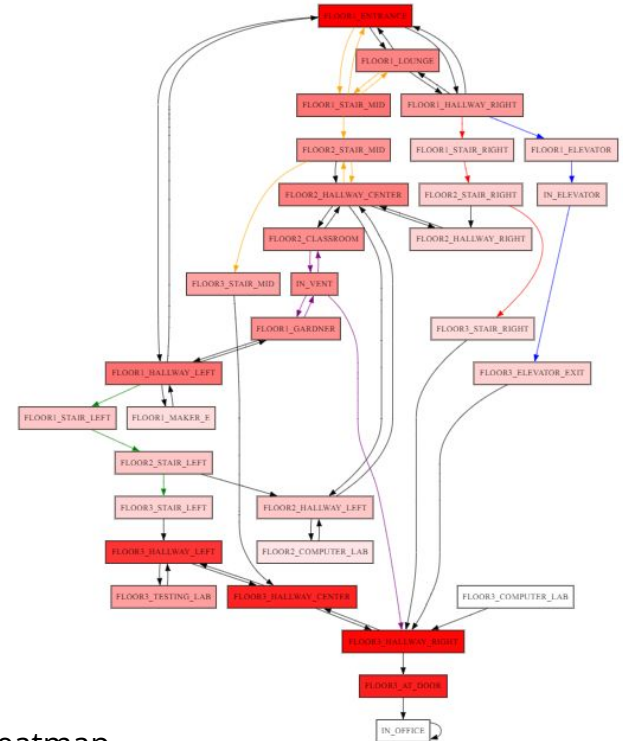
- 1 in 1,000,000 chance when checking the Maker-E camera, Instant AI type appears
- When the camera is closed, the player will instantly lose the game



Student AI Implementation

Weighted Directed Graph

- A graph where edges have a specific direction—pointing from one vertex to another—and an assigned numerical value (weight) representing cost, distance, or capacity
- Each location is a vertex
- Each legal move is an edge
- Each edge has a weight representing how attractive the move is
- On each decision step outgoing edge weights are adjusted by discrete factors (personality, floor progression, backtrack penalties, recent memory) then normalized

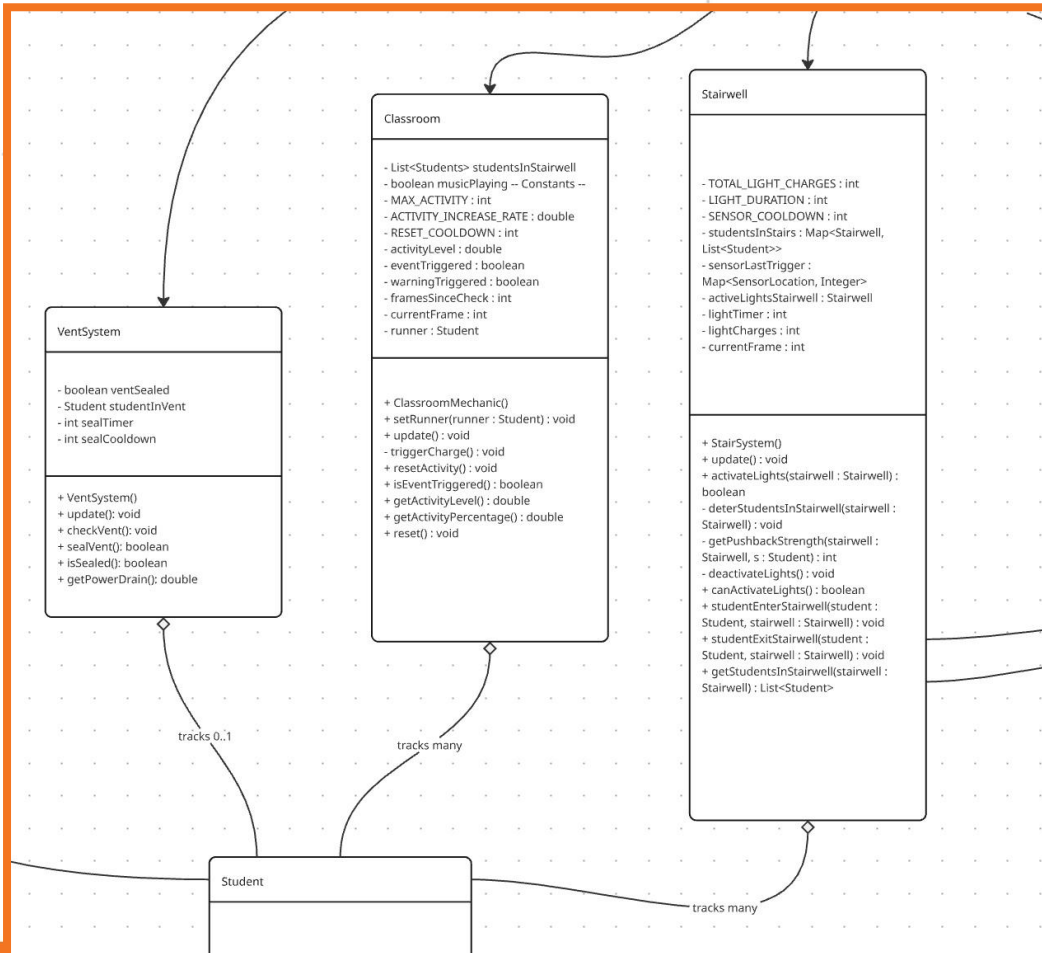


SHY Heatmap

Code Design

Separation of Responsibility

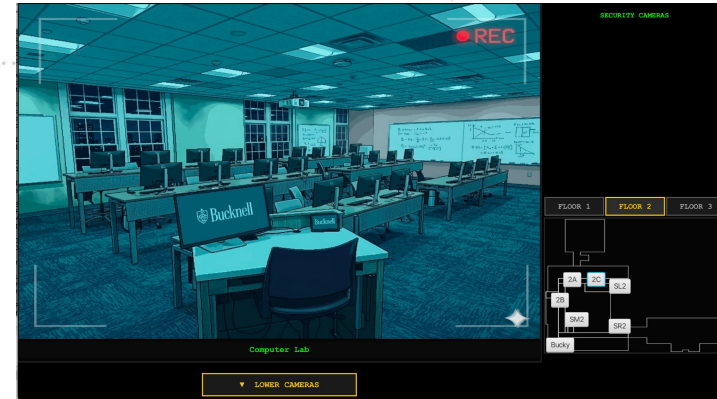
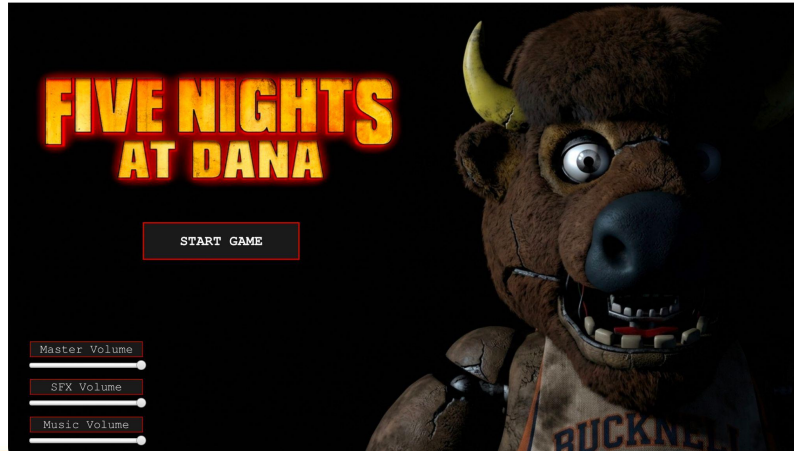
- Our system follows the principle of separation of responsibility by assigning distinct tasks to dedicated classes.
- **AI Layer**
 - Models student behavior and movement logic
- **Managers**
 - Coordinate cross-system rules and shared states
- **Controllers**
 - Handle user input and UI flow
- **Systems**
 - Implement gameplay mechanics



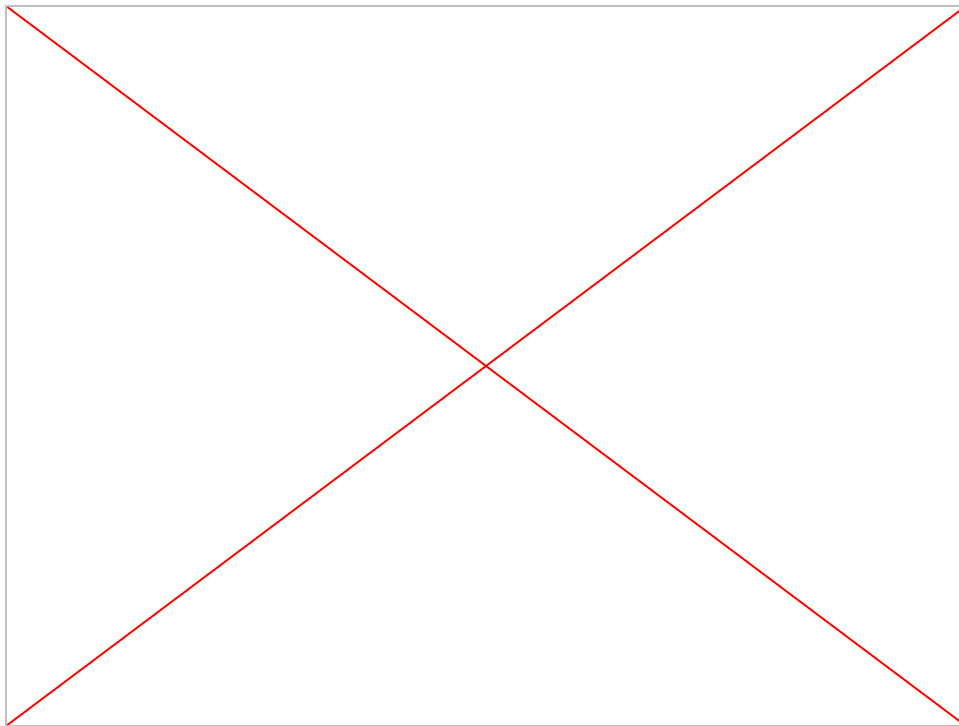
UI Design and Visuals

Scenebuilder:

- Our system uses javafx scene builder to develop all the UI
- Additional AI generated art was generated to give our game visual feedback for gameplay
- This art would be replaced in the long run with final assets



Demo



Scrum Process and Implementation

Team Dynamics and Scrum

- **Scrum Execution:** The project utilized a 4 sprint Agile approach. The team started slow in Sprint 1 due to onboarding but was going strong by Sprint 3.
- **Roles:** Roles stayed constant throughout the project. Work was distributed based on independent tasks.
- **Challenges:** We found that everyone doing work at the start is important as it ensures the whole group understands the core systems and if that doesn't happen it can be challenging to catch up.

Successes and Challenges

- **Major Success:** UML planning made merging complex systems like AI and cameras painless.
- **Implementation Hardships:** Game balancing was harder than expected. AI speeds and the energy mechanic was difficult to iron out within the timeframe.
- **Team Git Usage:** We successfully had everyone merging to dev and main frequently in later sprints to keep all members synchronized with current code.

Lessons Learned and the Future

Reflections and Improvements

- **What to Change:** If we restarted, we would narrow the scope (fewer floors and cameras) to focus more on game balance.
- **Workload Management:** A primary goal for future projects would be to equalize our workload distribution early to ensure all members grasp the full system architecture.
- **Communication:** We would also prioritize making more tasks and having very clear guidelines. The clearer the better.

Time Estimates and Future Impact

- **Estimating Progress:** Our initial estimates were off during Sprint 1, but we improved by Week 2 as the team aligned with the ideal burndown slope and knew what needed to be done.
- **Total Effort:** The project concluded with 160 total hours.
- **We found this experience really emphasized the importance of early stage planning and UML so that we knew what needed to be done later on.**

Questions?